Model Predictive Formation Tracking-Containment Control for Multi-UAVs With Obstacle Avoidance

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***Abstract*—This article investigates the formation tracking-containment control problem for multiple unmanned aerial vehicles (UAVs), while considering collision avoidance in the three-dimensional (3-D) environment. A distributed model predictive formation control method is developed for UAVs with obstacle avoidance. Unlike most existing trajectory tracking control schemes, the proposed method is derived from a smooth shifting function and a distributed Lyapunov-based model predictive controller, together with two collision-free functions, to obtain an improved control algorithm with the following characteristics: 1) the swarm system of UAVs realizes the cooperative hunting process from tracking formation to containing formation, fencing a target to their convex hull; 2) the Lyapunov-based model predictive formation control method inherits the stability properties of backstepping technique and adopts the receding horizon optimization of model predictive control (MPC) technique; and 3) by exploiting the Lyapunov-based MPC algorithm and collision-free functions, the swarm system of UAVs can avoid collision with obstacles or collision with each other in 3-D scenarios with multiple obstacles. Finally, the simulation and comparison results confirm that the proposed controller outperforms the traditional backstepping controller in terms of safety and tracking performance.**

***Index Terms*—Lyapunov-based model predictive control (MPC), obstacle avoidance, shifting function, tracking-containment control, unmanned aerial vehicles (UAVs).**

I. Introduction

NOWADAYS, quadrotor unmanned aerial vehicle (UAV) has become the mainstream of small UAV due to its advantages of easy operation, high reliability, high agility, and so on [1], [2], [3]. By controlling the speed of each propeller, the quadrotor UAV can quickly switch between different flight modes. In particular, multiple UAVs formation control, which aims to control a group of UAVs to complete a given task through local communication, draws the attention of the researchers due to its potential applications in the fields of search, rescue and cluster battles. Some primary efforts have been dedicated to achieve the formation control of quadrotor UAVs, including sliding mode control [4], PID control [5], model predictive control (MPC) [6], and backstepping control [7]. However, the desired control performance is often not achieved using a single control approach. Therefore, it is significant to develop improved control schemes to optimize control performance.

MPC, with its ability to solve multivariate constrained problems while considering optimization factors, has attracted wide attention. The main idea of MPC is to select control actions by iteratively solving online constrained optimization problem within a finite prediction horizon. Taking the linearized dynamics of system states as the predictive model, Berberich et al. [8] investigated the consensus tracking control problem for nonlinear systems. By combining the image kinematics and quadrotor dynamic, a robust anti-disturbance MPC approach was designed for the visual servoing of UAVs [9]. It is worth noting that MPC strategies use a finite prediction horizon, so the solution's optimality does not guarantee the stability of the closed-loop system. The traditional MPC strategies use infinite prediction horizons or add a terminal penalty term in the cost function to achieve closed-loop system stability [8], [9], [10], [11]. However, the implicit nature of MPC control law brings great challenges to the closed-loop stability analysis. To this end, Lyapunov-based MPC (LMPC) paves the way to tackle such an intractable problem. The closed-loop stability and feasibility of LMPC are guaranteed by Lyapunov-based nonlinear control laws. For example, taking data losses into consideration, Christofides et al. [12] proposed a LMPC algorithm for the networked control system, in which the nonlinear controller was obtained by Artstein's theorem. In [13], a LMPC protocol was proposed for the attitude system for quadrotor UAV, where the nonlinear controller was obtained by a sliding mode control method. In [14], a LMPC scheme was proposed for underwater vehicles, and the stability and feasibility of the underwater vehicle control system were guaranteed by a backstepping controller. It is worth noting that the above LMPC methods have the advantage of constrained optimization of MPC while inheriting the stability properties of the Lyapunov-based nonlinear controller. However,

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the aforementioned results [12], [13], [14] only apply to the tracking task, and the agent unit follows the trajectory curve of a fixed distance, which is not applicable to the tracking-containment control of UAVs.

In the context of animal hunting in nature, swarm cooperative control aims to guide followers to enclose a specified target within a convex hull formed by the followers. In an encirclement mission of UAVs, a group of UAVs is required to surround a static or dynamic target in order to achieve surveillance and capture of the enemy target. In [15] and [16], the time-varying formation pursuit problem was studied by utilizing switching interaction topologies. In [17], a target encirclement problem of multi-UAVs was investigated. Instead of using the traditional approach of UAVs converging to a closed curve, some Lyapunov-like functions and an annular virtual tube were cleverly designed in [17] to make the UAVs converge to an annular virtual tube for target encirclement. In [18] and [19], two fencing control strategies for a moving target were proposed, where the controllers implement fenced by the vehicles based on the constantly changing topologies of neighbors. By introducing a switch function, Yu et al. [20] investigated the adaptive tracking-containment problem for a swarm system. It should be noted that transient switching may cause the collapse of the communication topology. Furthermore, the following UAVs must not only reactively avoid collisions, but also predictively adapt their positions to track and contain the moving target stably. Therefore, it is important to investigate the problem of optimization-based containment hunting for UAVs with a smoother fencing process.

When a UAV senses an obstacle, it must be able to avoid the obstacle to prevent the UAV from crashing and mission failure. As one of the fundamental problems of UAV, the obstacle avoidance technique has been extensively studied [21], [22], [23], [24], [25]. In [26], an adaptive navigation control strategy was proposed for robot trajectory planning with collision avoidance. In [27], a LMPC strategy, which adds a cost term based on artificial potential fields, was proposed to avoid collision avoidance among underwater vehicles. A trajectory tracking control algorithm was proposed in [28] by MPC scheme, which can avoid collision between UAV and obstacle. However, the results [26], [27], [28] only consider a single obstacle, or the obstacle avoidance between agents depends on the topology communication of relatively centralized control. To our best knowledge, few has adopted the LMPC framework to investigate the swarm cooperative hunting problem for UAVs in scenarios with multiple obstacles, which motivates this work.

In this article, a Lyapunov-based model predictive formation tracking-containment control technique is proposed for UAVs with obstacle avoidance. During the initial phase, the swarm system of UAVs is required to track a specified target. Once the preset time is reached, the swarm system of UAVs switches from tracking control to containment control, fencing the target to their convex hull. The relevant contributions are shown below.

1) Inspired by animal hunting behavior, a Lyapunov-based model predictive formation tracking-containment control technique is proposed. The key to achieving the cooperative hunting process is to construct a smooth transition function. The new cooperative hunting control method is based on the original communication topology rather than topology switching, which effectively avoids the collapse of topology communication caused by topology switching.
2) A comprehensive collision avoidance algorithm is proposed to achieve the swarm system of UAVs to safely navigate around obstacles. Compared with the existing obstacle avoidance results [23], [24], [26], [27], [28], our results can be applied in 3-D scene. Specifically, by incorporating two collision avoidance functions into the cost index, the comprehensive collision avoidance algorithm simultaneously avoids collisions between inter-UAVs and collisions between UAVs and obstacles. Then, the LMPC controller finds the tradeoff between formation error, system input, obstacle avoidance and collision avoidance.
3) To guarantee the closed-loop system's stability, this article adds an auxiliary controller $\boldsymbol{\eta}_i$ based on backstepping design as a constraint in the LMPC algorithm, instead of using the terminal penalty function in [8], [9], [10], and [11]. In doing so, the proposed LMPC inherits the stability properties of the auxiliary controller $\boldsymbol{\eta}_i$ and seeks the optimal solution within the feasible solution range of $\boldsymbol{\eta}_i$.

This article is organized as follows. Section II introduces the model of quadrotor UAV and the problem formulation. Section III presents the design of the LMPC algorithm in scenarios with multiple obstacles. The analysis of feasibility and stability is provided in Section IV. Finally, simulation and conclusions are drawn in Sections V and VI, respectively.

Notation: $\|\cdot\|_2$ and $\|\cdot\|_\infty$ denote the Euclidean norm and the infinity norm of a square matrix, respectively. $\max(\cdot)$ is the maximum value, $(\cdot)^{\mathrm{T}}$ represents the matrix transpose, and $\mathrm{diag}\{\cdot\}$ represents the diagonal matrix.

II. Preliminaries

A. Graph Theory

Consider $N+1$ UAVs consisting of N tracking UAVs and one tracked UAV. The communication topology can be described as $\mathcal{G} = (\mathcal{N}, \mathcal{E}, \boldsymbol{A})$, where $\mathcal{E} \in \mathcal{N} \times \mathcal{N}$ is the directed edges, and $\mathcal{N} = (1, \ldots, N)$ is the node set. Here, nodes $i (i = 1, 2, \ldots, N)$ denote tracking UAVs. $(\mathcal{N}_j, \mathcal{N}_i) \in \mathcal{E}$ denotes a directed edge from j to i. $N_i = \{\mathcal{N}_j \mid (\mathcal{N}_j, \mathcal{N}_i) \in \mathcal{E}, i \neq j\}$ denotes the set of neighbors of ith UAV. $\boldsymbol{A} = [a_{i,j}] \in \mathbb{R}^{N\times N}$ is the adjacency matrix. The information transmission from node j to node i with $a_{i,j} > 0$ if $(j, i) \in \mathcal{E}$, otherwise $a_{i,j} = 0$. The Laplacian matrix is $\bar{\boldsymbol{\Upsilon}} = \boldsymbol{L} - \boldsymbol{A}$, where $\boldsymbol{L} = \mathrm{diag}\{d_1, \ldots, d_N\} \in \mathbb{R}^{N\times N}$ is degree matrix with $d_i = \sum_{j=1}^{N} a_{i,j}$. The node $N+1$ denotes the tracked UAV. $\bar{\mathcal{G}} = (\bar{\mathcal{N}}, \bar{\mathcal{E}})$ is the augmented graph with $\bar{\mathcal{N}} = (1, \ldots, N+1)$ and $\bar{\mathcal{E}} \in \bar{\mathcal{N}} \times \bar{\mathcal{N}}$. $\bar{\boldsymbol{\mathcal{B}}} = \mathrm{diag}\{b_1, \ldots, b_N\} \in \mathbb{R}^{N\times N}$ is a diagonal matrix, where $b_i > 0$ if ith UAV can get information from tracked UAV, otherwise $b_i = 0$. A digraph graph contains a spanning tree, if there

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exists a tracked UAV, such that this tracked UAV has directed paths to all other tracking UAVs.

B. Model of Quadrotor UAV

The quadcopter UAV consists of a frame, a power system and a command system. The autopilot in the command system can control the stability of the system through the pilot's command. For the ith UAV, the kinematic equation is described as [6], [29], [30], [31]

$$\begin{cases} \dot{\boldsymbol{p}}_i = \boldsymbol{R}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i \\ \dot{\boldsymbol{\mathcal{V}}}_i = \boldsymbol{\beta}_i\boldsymbol{\mathcal{V}}_i + \boldsymbol{\gamma}_i\boldsymbol{\tau}_i \end{cases} \tag{1}$$

where $i = 1, 2, \ldots, N$, $\boldsymbol{p}_i = [x_i, y_i, z_i, \psi_i]^{\mathrm{T}} \in \mathbb{R}^4$ represents the position vector (transverse, longitudinal, vertical) and yaw angle, $\boldsymbol{\mathcal{V}}_i = [\vartheta_{ix}, \vartheta_{iy}, \vartheta_{iz}, \vartheta_{i\psi}]^{\mathrm{T}} \in \mathbb{R}^4$ with ϑ_{ix}, ϑ_{iy}, and ϑ_{iz} being the UAV linear speeds in the body frame and $\vartheta_{i\psi}$ being the heading angular speed. Its control input can be represented as $\boldsymbol{\tau}_i = [\tau_{ix}, \tau_{iy}, \tau_{iz}, \tau_{i\psi}]^{\mathrm{T}} \in \mathbb{R}^4$, where τ_{ix} is forward speed command, τ_{iy} is sidewards speed command, τ_{iz} is upward speed command, and $\tau_{i\psi}$ is yaw speed command. $\boldsymbol{\beta_i} = \mathrm{diag}(-[1/L_{ix}], -[1/L_{iy}], -[1/L_{iz}], -[1/L_{i\psi}]) \in \mathbb{R}^{4\times4}$. $\boldsymbol{\gamma_i} = \mathrm{diag}([K_{ix}/L_{ix}], [K_{iy}/L_{iy}], [K_{iz}/L_{iz}], [K_{i\psi}/L_{i\psi}]) \in \mathbb{R}^{4\times4}$, where K_{ia} and L_{ia} are the gain and the delay associated with ϑ_{ia} and τ_{ia} $(a = x, y, z, \psi)$. The physical input constraint of UAV need to satisfy $\|\boldsymbol{\tau}_i\|_\infty \le \tau_{\max}$, where $\tau_{\max}$ is the maximum allowable input for the UAV. $\boldsymbol{R}_{i,b}^{i,e}$ is the translation matrix from the body-fixed frame to the earth-fixed frame, which can be expressed as

$$\boldsymbol{R}_{i,b}^{i,e} = \begin{bmatrix} \cos\psi_i & -\sin\psi_i & 0 & 0 \\ \sin\psi_i & \cos\psi_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

C. Problem Formulation

To facilitate the subsequent analysis, the system (1) can be transformed into

$$\begin{aligned} \dot{\boldsymbol{\mathcal{X}}}_i &= \begin{bmatrix} \dot{\boldsymbol{\mathcal{X}}}_{i,1} \\ \dot{\boldsymbol{\mathcal{X}}}_{i,2} \end{bmatrix} = \begin{bmatrix} \dot{\boldsymbol{p}}_i \\ \dot{\boldsymbol{\mathcal{V}}}_i \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i \\ \boldsymbol{\beta}_i\boldsymbol{\mathcal{V}}_i + \boldsymbol{\gamma}_i\boldsymbol{\tau}_i \end{bmatrix} \\ \boldsymbol{\mathcal{Y}}_i &= \boldsymbol{\mathcal{X}}_{i,1} \end{aligned} \tag{3}$$

where $i = 1, 2, \ldots, N$, $\boldsymbol{\mathcal{X}}_{i,1}$ and $\boldsymbol{\mathcal{X}}_{i,2}$ are state vectors, $\boldsymbol{\mathcal{X}}_{i,1} = [x_i, y_i, z_i, \psi_i]^{\mathrm{T}} \in \mathbb{R}^4$, $\boldsymbol{\mathcal{X}}_{i,2} = [\vartheta_{ix}, \vartheta_{iy}, \vartheta_{iz}, \vartheta_{i\psi}]^{\mathrm{T}} \in \mathbb{R}^4$, $\boldsymbol{\tau}_i$ and $\boldsymbol{\mathcal{Y}}_i = [\mathcal{Y}_{i,1}, \mathcal{Y}_{i,2}, \mathcal{Y}_{i,3}, \mathcal{Y}_{i,4}]^{\mathrm{T}} \in \mathbb{R}^4$ denote the input and output, respectively. The reference trajectory of the tracked UAV is defined as $\boldsymbol{\mathcal{Y}}_d = [\mathcal{Y}_{dx}, \mathcal{Y}_{dy}, \mathcal{Y}_{dz}, \mathcal{Y}_{d\psi}]^{\mathrm{T}} = [x_r, y_r, z_r, \psi_r]^{\mathrm{T}} \in \mathbb{R}^4$.

Let $\boldsymbol{\mathcal{X}}_i$ be the state vector. The kinematic equation in (3) can be abbreviated as

$$\dot{\boldsymbol{\mathcal{X}}}_i = \boldsymbol{g}_i(\boldsymbol{\mathcal{X}}_i, \boldsymbol{\tau}_i). \tag{4}$$

The control goal is to design a LMPC algorithm to ensure that the following.

1) When the preset shifting time $T_{i,a}$ is reached, the swarm system of UAVs shifts from the tracking formation to the containment formation.
2) The swarm system of UAVs can achieve obstacle avoidance between UAVs and obstacles as well as collision avoidance between neighboring UAVs.
3) The UAV physical input constraint is satisfied $\|\boldsymbol{\tau}_i\|_\infty \le \tau_{\max}$.

To achieve the above control goal, the following assumptions and lemmas will be used.

Assumption 1: The augmented graph $\bar{\mathcal{G}}$ contains a spanning tree with the tracked UAV rooted.

Assumption 2: The reference signals $x_r(t)$, $y_r(t)$ and $z_r(t)$ are 2th order continuous differentiable and bounded, i.e., the constant $\bar{\mathcal{Y}}_d$ is assumed to satisfy $\bar{\mathcal{Y}}_d \ge \max\{\|x_r(t)\|, \|y_r(t)\|, \|z_r(t)\|, \|\dot{x}_r(t)\|, \|\dot{y}_r(t)\|, \|\dot{z}_r(t)\|, \|\ddot{x}_r(t)\|, \|\ddot{y}_r(t)\|, \|\ddot{z}_r(t)\|\}$.

Lemma 1 [32]: Let Assumption 1 hold. Then, $\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}}$ is non singular.

For the tracked UAV reference signal, we define the following reference states:

$$\begin{cases} \psi_r(t) = \mathrm{atan2}\{\dot{y}_r(t), \dot{x}_r(t)\} \\ \vartheta_{rx}(t) = \sqrt{\dot{x}_r^2(t) + \dot{y}_r^2(t)} \\ \vartheta_{ry}(t) = 0 \\ \vartheta_{rz}(t) = \dot{z}_r(t) \\ \vartheta_{r\psi}(t) = \frac{\dot{x}_r\ddot{y}_r - \ddot{x}_r\dot{y}_r(t)}{\dot{x}_r^2(t) + \dot{y}_r^2(t)}. \end{cases} \tag{5}$$

The formation error is defined as

$$\begin{aligned} \boldsymbol{S}_{i,1} = &\sum_{j=1}^{N} a_{i,j}\big(\boldsymbol{\mathcal{Y}}_j - \boldsymbol{M}_j - \boldsymbol{G}_j - \boldsymbol{\mathcal{Y}}_i + \boldsymbol{M}_i + \boldsymbol{G}_i\big) \\ &+ b_i(\boldsymbol{\mathcal{Y}}_d - \boldsymbol{\mathcal{Y}}_i + \boldsymbol{M}_i + \boldsymbol{G}_i) \end{aligned} \tag{6}$$

where $\boldsymbol{S}_{i,1} = [S_{i,x1}, S_{i,y1}, S_{i,z1}, S_{i,\psi1}]^{\mathrm{T}} \in \mathbb{R}^4$. $\boldsymbol{M}_i$ and $\boldsymbol{M}_j$ are the relative distances of i and j with respect to the tracked UAV, respectively. $\boldsymbol{G}_i = [G_{ix}, G_{iy}, G_{iz}, 0]^{\mathrm{T}} \in \mathbb{R}^4$ and $\boldsymbol{G}_j = [G_{jx}, G_{jy}, G_{jz}, 0]^{\mathrm{T}} \in \mathbb{R}^4$ are smooth shifting functions. The transition functions from tracking formation to containment formation can be described as

$$\begin{cases} G_{ia} = \bar{h}_i \exp\left(-\dfrac{\left(\tan\left(\frac{\pi(T_{i,a}-t)}{2T_{i,a}}\right)\right)^4}{2\delta_i^4} \right), & 0 \le t < T_{i,a} \\ G_{ia} = \bar{h}_i, & t > T_{i,a} \end{cases} \tag{7}$$

$$\begin{cases} G_{ja} = \bar{h}_j \exp\left(-\dfrac{\left(\tan\left(\frac{\pi(T_{j,a}-t)}{2T_{j,a}}\right)\right)^4}{2\delta_j^4} \right), & 0 \le t < T_{j,a} \\ G_{ja} = \bar{h}_j, & t > T_{j,a} \end{cases} \tag{8}$$

where $a = x, y, z$, δ_i and δ_j are the rate parameters, $T_{i,a}$ and $T_{j,a}$ are the preset shifting times, $\bar{h}_i$ and $\bar{h}_j$ are negative parameters.

Remark 1: For the formation error function (6), the shifting functions $\boldsymbol{G}_i$ and $\boldsymbol{G}_j$ are added to the ith UAV and jth UAV, respectively. By introducing $\boldsymbol{G}_i$ and $\boldsymbol{G}_j$, the swarm system of UAVs achieves the shifting process from tracking formation to containment formation. In the shifting functions (7) and (8), the design parameters $\bar{h}_i$ and $\bar{h}_j$ represent the adjustment variables of formation distance, and δ_i and δ_j control the convergence rate. The larger the values of δ_i and δ_j, the slower the transition from tracking control to containment control.

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TABLE I
OBSTACLE AVOIDANCE PARAMETERS

Parameters	Values
Minimum safe distance	$\mathcal{L}_l$
Detection range	$\bar{D}_{il}$
Distance between the ith UAV and the lth obstacle	D_{il}
Obstacle avoidance weight parameters	$\mathcal{M}_{il}$

Remark 2: In [17], the UAVs are restricted within an annular virtual tube, aiming to achieve approximately even encirclement based on the barrier function approach, given a generic closed generating curve. Instead of using the traditional approach of UAVs converging to a closed curve, the curve virtual tube encirclement control is an innovative approach. Different from [17] which acquires other UAV information through onboard detection devices, this article obtains other UAV information based on the communication topology. The method proposed utilizes a shifting function based on the original topology communication information to achieves the shifting process from tracking formation to containment formation. For model predictive encirclement control of UAVs within a virtual tube, it needs further study.

Lemma 2 [33]: For $\mathbf{Y}_{da} = (\mathcal{Y}_{da}, \ldots, \mathcal{Y}_{da})^{\mathrm{T}} \in \mathbb{R}^N$, $\boldsymbol{S}_{\bullet,a1} = (S_{1,a1}, \ldots, S_{N,a1})^{\mathrm{T}} \in \mathbb{R}^N$ and $\boldsymbol{\mathcal{Y}}_a = (\mathcal{Y}_{1,a}, \ldots, \mathcal{Y}_{N,a})^{\mathrm{T}} \in \mathbb{R}^N$, one has

$$\|\boldsymbol{\mathcal{Y}}_a - \mathbf{Y}_{da}\| \leq \|\boldsymbol{S}_{\bullet,a1}\| / \Delta\left(\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}}\right) \tag{9}$$

where $a = x, y, z, \psi$, $\Delta(\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}})$ is the minimum singular value of $\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}}$.

D. Obstacle Avoidance and Collision Avoidance

The swarm system of UAVs is often carried out in complex environments with multiple obstacles. To achieve inter-UAV collision avoidance and UAV-obstacle avoidance, a comprehensive collision and obstacle avoidance algorithm is introduced in the cost index: 1) a collision avoidance function that addresses inter-UAV collision avoidance and 2) an obstacle avoidance function that addresses UAV-obstacle avoidance.

The obstacle avoidance function between UAV and obstacles can be expressed as

$$H_{il}^{oa} = \max\{-\mathcal{M}_{il}\mathrm{In}\left(\frac{\|D_{il}\| - \mathcal{L}_l}{\bar{D}_{il} - \mathcal{L}_l}\right), 0\} \tag{10}$$

where l represents the lth obstacle, the definitions of obstacle avoidance parameters are listed in Table I. $D_{il} = (\sqrt{(x_i - X_{lo})^2 + (y_i - Y_{lo})^2 + (z_i - Z_{lo})^2})$ represents the Euclidean distance between the ith UAV and the lth obstacle in 3-D coordinates, where $[X_{lo}, Y_{lo}, Z_{lo}]$ is the position coordinate of the lth obstacle (transverse, longitudinal, vertical). When the distance between the ith UAV and the lth obstacle is greater than $\bar{D}_{il}$, the potential of the obstacle avoidance function is 0. At this time, the UAV does not need to consider obstacle avoidance. When the distance between the ith UAV and the lth obstacle is less than $\bar{D}_{il}$, the potential of the obstacle avoidance function rises rapidly.

TABLE II
COLLISION AVOIDANCE PARAMETERS

Parameters	Values
Minimum safe distance	$\bar{\mathcal{D}}_{il}$
Distance between the ith UAV and the neighboring UAV	$\mathcal{D}_{il}$
Collision avoidance weight parameters	$\wp_{ij}$

The inter-UAVs collision avoidance function can be expressed as

$$\Re_{ij}^{ca} = \sum_{j=1}^{N} a_{i,j} \frac{\wp_{ij}}{1 + \exp\left(||\mathcal{D}_{ij}|| - \bar{\mathcal{D}}_{ij}\right)} \tag{11}$$

where j represents the jth UAV with $i \neq j$, the definitions of collision avoidance parameters are listed in Table II. $\mathcal{D}_{ij} = (\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2})$ represents the Euclidean distance between the ith UAV and the jth UAV in 3-D coordinates.The LMPC controller finds the tradeoff between formation error, system input, obstacle avoidance and collision avoidance.

Remark 3: The proposed strategy can systematically solve the obstacle avoidance problem, which is often ignored in many existing results, especially for traditional nonlinear controllers, such as backstepping controllers and sliding mode controllers [4], [5], [6], [7]. The LMPC controller finds the tradeoff between formation error, system input, obstacle avoidance and collision avoidance, while maintaining system stability.

III. LYAPUNOV-BASED MODEL PREDICTIVE CONTROLLER DESIGN

In this section, we present the LMPC algorithm for the swarm system of UAVs with collision avoidance. Utilizing the predictive minimization cost index obtained from the kinematics of UAV, the Lyapunov-based model predictive controller obtains the optimized control input by repeatedly solving an online constrained optimization problem within a finite prediction horizon.

A. Design of LMPC

The LMPC strategy uses receding horizon optimization to find the tradeoff between formation error, system input, and obstacle avoidance. By using the sequential quadratic programming method, the optimal control signal $\boldsymbol{\tau}_i$ of the ith UAV can be found by minimizing the following cost index:

$$\begin{aligned} J_i = \min_{\hat{\boldsymbol{\tau}}_i \in \zeta(\Delta)} \int_{t_k}^{t_k+T} \Bigg\{ \sum_{j=1}^{N} \Big[&\|\hat{\boldsymbol{S}}_{i,1}(\varrho; t_k)\|_{\boldsymbol{Q}_J}^2 \\ &+ H_{il}^{oa}\left(\hat{\boldsymbol{\mathcal{X}}}_i(\varrho; t_k)\right) + \Re_{ij}^{ca}\left(\hat{\boldsymbol{\mathcal{X}}}_{\boldsymbol{i}}(\varrho; t_k), \hat{\boldsymbol{\mathcal{X}}}_{\boldsymbol{j}}(\varrho; t_k)\right) \Big] \\ &+ \|\hat{\boldsymbol{\tau}}_i(\varrho; t_k)\|_{\boldsymbol{F}_J}^2 \Bigg\} ds \end{aligned} \tag{12}$$

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subject to

$$\dot{\hat{\mathcal{X}}}_i(\varrho;t_k)=\boldsymbol{g}_i\big(\hat{\mathcal{X}}_i(\varrho;t_k),\hat{\tau}_i(\varrho;t_k)\big) \tag{13}$$

$$\hat{\boldsymbol{\mathcal{X}}}_i(t_k;t_k)=\boldsymbol{\mathcal{X}}_i(t_k) \tag{14}$$

$$\|\hat{\boldsymbol{\tau}}_i(\varrho;t_k)\|_\infty\le\tau_{\max} \tag{15}$$

$$\frac{\partial V_{i,2}}{\partial\boldsymbol{\mathcal{X}}_i}\boldsymbol{g}_i\Big(\hat{\boldsymbol{\mathcal{X}}}_i(t_k;t_k),\hat{\boldsymbol{\tau}}_i(t_k;\hat{\boldsymbol{\mathcal{X}}}_i(t_k;t_k))\Big)\le\frac{\partial V_{i,2}}{\partial\boldsymbol{\mathcal{X}}_i}\boldsymbol{g}_i\Big(\hat{\boldsymbol{\mathcal{X}}}_i(t_k;t_k),\boldsymbol{\eta}_i\big(\hat{\boldsymbol{\mathcal{X}}}_i(t_k)\big)\Big) \tag{16}$$

where t_k is the current time instant, $\varrho\in[t_k,t_k+T]$, $T=\aleph\Delta$ represents the prediction horizon, T is divided into $\aleph$ steps and Δ is the sampling period with $t_{k+1}=t_k+\Delta$, $\zeta(\Delta)$ is a family of piece-wise constant functions, $\hat{\boldsymbol{\mathcal{X}}}_i$ and $\hat{\boldsymbol{\mathcal{X}}}_j$ are the predicted states, $\hat{\boldsymbol{S}}_{i,1}$ represents the predicted formation error with $\hat{\boldsymbol{S}}_{i,1}=\sum_{j=1}^{N}a_{i,j}(\hat{\boldsymbol{\mathcal{Y}}}_j-\boldsymbol{M}_j-\hat{\boldsymbol{G}}_j-\hat{\boldsymbol{\mathcal{Y}}}_i+\boldsymbol{M}_i+\hat{\boldsymbol{G}}_i)+b_i(\boldsymbol{\mathcal{Y}}_d-\hat{\boldsymbol{\mathcal{Y}}}_i+\boldsymbol{M}_i+\hat{\boldsymbol{G}}_i)$, $\hat{\boldsymbol{\tau}}_i$ is the predicted control input, the initial state at time t_k is $\boldsymbol{\mathcal{X}}_i(t_k)$. $\boldsymbol{\eta}_i$ stands for nonlinear auxiliary controller obtained by Lyapunov stability theorem with $\boldsymbol{\eta}_i=[\eta_{i,x},\eta_{i,y},\eta_{i,z},\eta_{i,\psi}]^{\mathrm{T}}\in\mathbb{R}^4$. $\boldsymbol{Q}_J\in\mathbb{R}^{4\times4}$ is the symmetric weighted matrix of formation errors, $\boldsymbol{F}_J\in\mathbb{R}^{4\times4}$ is the symmetric weighted matrix for the system input, $H_{il}^{oa}(\hat{\boldsymbol{\mathcal{X}}}_{\boldsymbol{i}}(\varrho;t_k))$ represents the obstacle avoidance function between UAV and obstacle, $\Re_{ij}^{ca}(\hat{\boldsymbol{\mathcal{X}}}_{\boldsymbol{i}}(\varrho;t_k),\hat{\boldsymbol{\mathcal{X}}}_{\boldsymbol{j}}(\varrho;t_k))$ represents the collision avoidance function between UAVs, $V_{i,2}$ is the candidate Lyapunov function. The optimal solution to this LMPC problem can be expressed as ${\boldsymbol{\tau}^*}_i(\varrho,t_k)$.

In (13)–(16), (13) represents the dynamic constraints of the kinematic (1) to predict future trajectories. Constraint (14) is the constraint on the control input. Constraint (15) provides the initial state for the LMPC at time t_k. Note that the receding horizon optimization of MPC, within a finite prediction horizon, does not guarantee the stability and feasibility of the closed-loop system. To ensure closed-loop stability, an additional constraint (16) is added based on the Lyapunov-based nonlinear auxiliary controller $\boldsymbol{\eta}_i$ derived via backstepping control. Backstepping control is a recursive method. It breaks down the control problem into smaller subproblems and uses a recursive algorithm to construct a series of Lyapunov functions to guarantee system stability [34]. The design of the backstepping-based auxiliary controller $\boldsymbol{\eta}_i$ is provided in Section III-B.

Constraint (16) guarantees that the derivative of Lyapunov function $V_{i,2}(\boldsymbol{\mathcal{X}}_i)$ at the moment t_k obtained by the Lyapunov-based model predictive controller ${\boldsymbol{\tau}^*}_i$ is smaller than or equal to the derivative by the actual auxiliary controller $\boldsymbol{\eta}_i$.

Remark 4: The traditional MPC strategies need to add a terminal penalty into the cost index to ensure closed-loop stability [8], [9], [10], [11]. However, for complex nonlinear systems, the implicit nature of nonlinear MPC control law makes the stability region difficult to explicitly characterize. In contrast, the designed LMPC algorithm achieves the separation between system stability and performance optimization. By introducing (16), the LMPC algorithm inherits the stability and robustness properties of the nonlinear control law $\boldsymbol{\eta}_i$ and seeks the optimal solution within the feasible solution range of $\boldsymbol{\eta}_i$. This explicit indication of the stability region of attraction allows the proposed LMPC strategy to separate system stability from performance optimization without requiring the addition of a terminal penalty term in the cost function [8], [9], [10], [11].

Algorithm 1: LMPC

Input: For each UAV, input the cost index J_i (12) with $i=1,2,\ldots,N$;

Initialize: For the swarm system of UAVs, set the initial state $\boldsymbol{\mathcal{X}}_i(0)$ of each UAV, the simulation step T_S, the prediction horizon T, the sampling period Δ and the weight matrix $\boldsymbol{Q}_J$ and $\boldsymbol{F}_J$, $\tau_{\max}$ is the maximum input limit, the adjustment parameters $\mathcal{M}_{il}$, $\mathcal{L}_l$, $\wp_{ij}$, $\bar{D}_{il}$ and $\bar{\mathcal{D}}_{i,j}$; Set $k=0$;

The formation errors between UAVs and tracked UAV are set. For tracked UAV, set its flight trajectory.

Repeat:

for $k\le T_S$ **do**

- The ith UAV samples its system state $\boldsymbol{\mathcal{X}}_i$.
- The ith UAV captures the state information of tracked UAV or neighboring UAVs.
- Using the backstepping control strategy, (16) is obtained based on the actual auxiliary controller $\boldsymbol{\eta}_i$.
- Solve the cost index J_i (12) and generate the suboptimal solution ${\boldsymbol{\tau}^*}_i(\varrho,t_k)$ for $\varrho\in[t_k,t_k+T]$.
- Implement ${\boldsymbol{\tau}^*}_i$ for only one sampling period: $\boldsymbol{\tau}_i(\varrho;t_k)={\boldsymbol{\tau}^*}_i(\varrho;t_k)$ for $\varrho\in[t_k,t_k+\Delta]$.
- $k=k+1$, $t_{k+1}=t_k+\Delta$.

End

Remark 5: The traditional adaptive nonlinear control methods, including backstepping control method and sliding mode control method, need to consider the input constraints separately in the controller design process. The LMPC method can obtain the optimal solution with explicit consideration of input constraints.

Remark 6: For the closed-loop system, the length of the horizon T does not affect the stability, but it does affect the control performance and computational efficiency of the closed-loop system.

B. Adaptive Auxiliary Controller Design

In this section, we use the backstepping control strategy to calculate the nonlinear auxiliary controller $\boldsymbol{\eta}_i$, as discussed below.

According to (6), the coordinate transformation is introduced as follows:

$$\boldsymbol{S}_{i,1}=\sum_{j=1}^{N}a_{i,j}\big(\boldsymbol{\mathcal{X}}_{j,1}-\boldsymbol{M}_j-\boldsymbol{G}_j-\boldsymbol{\mathcal{X}}_{i,1}+\boldsymbol{M}_i+\boldsymbol{G}_i\big)+b_i\big(\boldsymbol{\mathcal{Y}}_d-\boldsymbol{\mathcal{X}}_{i,1}+\boldsymbol{M}_i+\boldsymbol{G}_i\big) \tag{17}$$

$$\boldsymbol{S}_{i,2}=\dot{\boldsymbol{S}}_{i,1}+\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} \tag{18}$$

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where $\boldsymbol{S}_{i,2} = [S_{i,x2}, S_{i,y2}, S_{i,z2}, S_{i,\psi 2}]^{\mathrm{T}} \in \mathbb{R}^4$, $\boldsymbol{k}_{i,1} = \mathrm{diag}\{k_{i,x1}, k_{i,y1}, k_{i,z1}, k_{i,\psi 1}\} \in \mathbb{R}^{4\times 4}$ with $k_{i,a1}(a = x, y, z, \psi)$ being the positive control constant.

The coordinate transformation in (18) is designed to make the error $\boldsymbol{S}_{i,1}$ stable. For (17), the derivative of $\boldsymbol{S}_{i,1}$ is

$$\begin{aligned}\dot{\boldsymbol{S}}_{i,1} &= \sum_{j=1}^{N} a_{i,j}\big(\dot{\boldsymbol{\mathcal{X}}}_{j,1} - \dot{\boldsymbol{M}}_j - \dot{\boldsymbol{G}}_j - \dot{\boldsymbol{\mathcal{X}}}_{i,1} + \dot{\boldsymbol{M}}_i + \dot{\boldsymbol{G}}_i\big) \\ &\quad + b_i\big(\dot{\boldsymbol{\mathcal{Y}}}_d - \dot{\boldsymbol{\mathcal{X}}}_{i,1} + \dot{\boldsymbol{M}}_i + \dot{\boldsymbol{G}}_i\big) \\ &= -(d_i + b_i)\Big(\boldsymbol{R}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i - \dot{\boldsymbol{G}}_i\Big) + \sum_{j\in N_i} a_{i,j} \\ &\quad \times \big(\boldsymbol{R}_{j,b}^{j,e}(\psi_j)\boldsymbol{\mathcal{V}}_j - \dot{\boldsymbol{G}}_j\big) + b_i\dot{\boldsymbol{\mathcal{Y}}}_d. \end{aligned} \tag{19}$$

Define the Lyapunov function candidate as

$$V_{i,1} = \frac{1}{2}\boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{S}_{i,1}. \tag{20}$$

Invoking (19) and (20), it follows that:

$$\begin{aligned}\dot{V}_{i,1} &= \boldsymbol{S}_{i,1}^{\mathrm{T}}\Big(-(d_i + b_i)\Big(\boldsymbol{R}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i - \dot{\boldsymbol{G}}_i\Big) \\ &\quad + \sum_{j\in N_i} a_{i,j}\Big(\boldsymbol{R}_{j,b}^{j,e}(\psi_j)\boldsymbol{\mathcal{V}}_j - \dot{\boldsymbol{G}}_j\Big) + b_i\dot{\boldsymbol{\mathcal{Y}}}_d\Big) \\ &\quad + \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} - \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} \\ &= \boldsymbol{S}_{i,1}^{\mathrm{T}}\big(\dot{\boldsymbol{S}}_{i,1} + \boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1}\big) - \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} \\ &= \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{S}_{i,2} - \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1}. \end{aligned} \tag{21}$$

Design a feedback control controller $\boldsymbol{\eta}_i$ to converge the error $\boldsymbol{S}_{i,2}$ of (18). Applying (19), the derivative of $\boldsymbol{S}_{i,2}$ is

$$\begin{aligned}\dot{\boldsymbol{S}}_{i,2} &= \ddot{\boldsymbol{S}}_{i,1} + \boldsymbol{k}_{i,1}\dot{\boldsymbol{S}}_{i,1} \\ &= -(d_i + b_i)\Big(\dot{\boldsymbol{R}}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i + \boldsymbol{R}_{i,b}^{i,e}(\psi_i)\dot{\boldsymbol{\mathcal{V}}}_i - \ddot{\boldsymbol{G}}_i\Big) \\ &\quad + \sum_{j\in N_i} a_{i,j}\Big(\dot{\boldsymbol{R}}_{j,b}^{j,e}(\psi_j)\boldsymbol{\mathcal{V}}_j + \boldsymbol{R}_{j,b}^{j,e}(\psi_j)\dot{\boldsymbol{\mathcal{V}}}_j - \ddot{\boldsymbol{G}}_j\Big) \\ &\quad + b_i\ddot{\boldsymbol{\mathcal{Y}}}_d + \boldsymbol{k}_{i,1}\big(\boldsymbol{S}_{i,2} - \boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1}\big). \end{aligned} \tag{22}$$

Consider the candidate Lyapunov function as

$$V_{i,2} = V_{i,1} + \frac{1}{2}\boldsymbol{S}_{i,2}^{\mathrm{T}}\boldsymbol{S}_{i,2}. \tag{23}$$

Under (22) and (23), calculating the derivative of $V_{i,2}$ yields

$$\begin{aligned}\dot{V}_{i,2} &= \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{S}_{i,2} - \boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} + \boldsymbol{S}_{i,2}^{\mathrm{T}}\Big(-(d_i + b_i) \\ &\quad \times \Big(\dot{\boldsymbol{R}}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i + \boldsymbol{R}_{i,b}^{i,e}(\psi_i)\big(\boldsymbol{\beta}_i\boldsymbol{\mathcal{V}}_i + \boldsymbol{\gamma}_i\boldsymbol{\eta}_i\big) - \ddot{\boldsymbol{G}}_i\Big) \\ &\quad + \sum_{j\in N_i} a_{i,j}\Big(\dot{\boldsymbol{R}}_{j,b}^{j,e}(\psi_j)\boldsymbol{\mathcal{V}}_j + \boldsymbol{R}_{j,b}^{j,e}(\psi_j)\dot{\boldsymbol{\mathcal{V}}}_j - \ddot{\boldsymbol{G}}_j\Big) \\ &\quad + \boldsymbol{k}_{i,1}\big(\boldsymbol{S}_{i,2} - \boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1}\big) + b_i\ddot{\boldsymbol{\mathcal{Y}}}_d\Big). \end{aligned} \tag{24}$$

The actual auxiliary controller $\boldsymbol{\eta}_i$ is designed as

$$\begin{aligned}\boldsymbol{\eta}_i &= -\boldsymbol{\gamma}_i^{-1}\boldsymbol{\beta}_i\boldsymbol{\mathcal{V}}_i + \frac{\boldsymbol{\gamma}_i^{-1}\Big(\boldsymbol{R}_{i,b}^{i,e}(\psi_i)\Big)^{\mathrm{T}}}{(d_i + b_i)}\Big((d_i + b_i)\ddot{\boldsymbol{G}}_i \\ &\quad + b_i\ddot{\boldsymbol{\mathcal{Y}}}_d - (d_i + b_i)\dot{\boldsymbol{R}}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i \\ &\quad + \sum_{j\in N_i} a_{i,j}\Big(\dot{\boldsymbol{R}}_{j,b}^{j,e}(\psi_j)\boldsymbol{\mathcal{V}}_j + \boldsymbol{R}_{j,b}^{j,e}(\psi_j)\dot{\boldsymbol{\mathcal{V}}}_j - \ddot{\boldsymbol{G}}_j\Big) \\ &\quad + \big(\boldsymbol{k}_{i,1} + \boldsymbol{k}_{i,2}\big)\boldsymbol{S}_{i,2} + \Big(1 - \boldsymbol{k}_{i,1}^2\Big)\boldsymbol{S}_{i,1}\Big) \end{aligned} \tag{25}$$

where $\boldsymbol{k}_{i,2} = \mathrm{diag}\{k_{i,x2}, k_{i,y2}, k_{i,z2}, k_{i,\psi 2}\} \in \mathbb{R}^{4\times 4}$ with $k_{i,a2}(a = x, y, z, \psi)$ being the positive control constant.

Based on (25), the derivative of $V_{i,2}$ can be written as

$$\dot{V}_{i,2} = -\boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} - \boldsymbol{S}_{i,2}^{\mathrm{T}}\boldsymbol{k}_{i,2}\boldsymbol{S}_{i,2}. \tag{26}$$

According to the Lyapunov stability theory, it follows from (26) that the ith UAV converges to the specified formation distance. Let $V_{i,\mathrm{LMPC}} = V_{i,2}$. The derivative of $V_{i,\mathrm{LMPC}}$ is

$$\begin{aligned}\dot{V}_{i,\mathrm{LMPC}} &= -\boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} + \boldsymbol{S}_{i,2}^{\mathrm{T}}\Big(\boldsymbol{\mathcal{H}}_i - \boldsymbol{R}_{i,b}^{i,e}\big(\boldsymbol{\beta}_i\boldsymbol{\mathcal{V}}_i + \boldsymbol{\gamma}_i\boldsymbol{\tau}_i\big)\Big) \\ &\le \dot{V}_{i,2} \end{aligned} \tag{27}$$

where $\boldsymbol{\mathcal{H}}_i = b_i\ddot{\boldsymbol{\mathcal{Y}}}_d - (d_i + b_i)\dot{\boldsymbol{R}}_{i,b}^{i,e}(\psi_i)\boldsymbol{\mathcal{V}}_i + (d_i + b_i)\ddot{\boldsymbol{G}}_i + \sum_{j\in N_i} a_{i,j}(\dot{\boldsymbol{R}}_{j,b}^{j,e}(\psi_j)\boldsymbol{\mathcal{V}}_j + \boldsymbol{R}_{j,b}^{j,e}(\psi_j)\dot{\boldsymbol{\mathcal{V}}}_j - \ddot{\boldsymbol{G}}_j) + \boldsymbol{k}_{i,1}\boldsymbol{S}_{i,2} + (1 - \boldsymbol{k}_{i,1}^2)\boldsymbol{S}_{i,1}$. The derivative of Lyapunov function $V_{i,\mathrm{LMPC}}(\boldsymbol{\mathcal{X}}_i)$ at the moment t_k obtained by the Lyapunov-based model predictive controller $\boldsymbol{\tau}_i$ is smaller than or equal to the derivative by the actual auxiliary controller $\boldsymbol{\eta}_i$. The proposed LMPC inherits the stability properties of the backstepping controller $\boldsymbol{\eta}_i$. Therefore, it indicates that the constraint (15) for the LMPC algorithm holds.

IV. Analysis of Feasibility and Stability

A. Recursive Feasibility Analysis

To guarantee the feasibility and the stability of the LMPC algorithm, the actual auxiliary controller must satisfy $||\boldsymbol{\eta}_i||_\infty \le \tau_{\max}$. The following lemma is introduced.

Lemma 3: For the actual auxiliary controller $\boldsymbol{\eta}_i$, the velocity $\boldsymbol{\mathcal{V}}_i$ satisfies the following conditions:

$$||\boldsymbol{\mathcal{V}}_i||_\infty \le \sqrt{2}\big(\|\boldsymbol{\mathcal{P}}\|_\infty + \bar{\mathcal{Y}}_d\big). \tag{28}$$

Proof: Consider the Lyapunov function candidate of UAVs as [35]

$$V = \sum_{i=1}^{N} V_{i,2}. \tag{29}$$

According to (26) and (29), we have

$$\begin{aligned}\dot{V} &\le \sum_{i=1}^{N}\big(-\boldsymbol{S}_{i,1}^{\mathrm{T}}\boldsymbol{k}_{i,1}\boldsymbol{S}_{i,1} - \boldsymbol{S}_{i,2}^{\mathrm{T}}\boldsymbol{k}_{i,2}\boldsymbol{S}_{i,2}\big) \\ &\le -\lambda V \end{aligned} \tag{30}$$

where $\lambda = \min\{2\|\boldsymbol{k}_{i,1}\|_{-\infty}, 2\|\boldsymbol{k}_{i,2}\|_{-\infty}\} > 0, i = 1, 2, \ldots, N$.

Multiplying $e^{\lambda t}$ from both sides of (30), we get

$$e^{\lambda t}\dot{V} \le -e^{\lambda t}\lambda V. \tag{31}$$

In view of the above analysis, one has

$$\frac{d\big(V(t)e^{\lambda t}\big)}{dt} \le 0. \tag{32}$$

Further, we have

$$V(t) \le V(0)e^{-\lambda t}. \tag{33}$$

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From (33), it follows that $\|\boldsymbol{S}_{i,1}(t_k)\|_\infty \leq \|\boldsymbol{S}_{i,1}(t_0)\|_2$ and $\|\boldsymbol{S}_{i,2}(t_k)\|_\infty \leq \|\boldsymbol{S}_{i,2}(t_0)\|_2$. Then, we derive from (18) that $\dot{\boldsymbol{S}}_{i,1}$ is bounded. ■

By applying Lemma 2 and (18), one has

$$\left\|\dot{\mathcal{Y}}_a - \dot{\mathbf{Y}}_{da}\right\|_\infty \leq \frac{\|\dot{\boldsymbol{S}}_{\bullet,a1}\|_2}{\Delta\left(\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}}\right)} \leq \frac{||\boldsymbol{S}_{\bullet,a2}(t_0)||_2 + k_{i,a1}||\boldsymbol{S}_{\bullet,a1}(t_0)||_2}{\Delta\left(\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}}\right)} \tag{34}$$

where $a = x, y, z, \psi$ and $\boldsymbol{S}_{\bullet,a2} = [S_{1,a2}, \ldots, S_{N,a2}]^{\mathrm{T}} \in \mathbb{R}^N$.

Let $\mathcal{P} = [\mathcal{P}_x, \mathcal{P}_y, \mathcal{P}_z, \mathcal{P}_\psi]^{\mathrm{T}} \in \mathbb{R}^4$, where $\mathcal{P}_a = ([||\boldsymbol{S}_{\bullet,a2}(t_0)||_2 + k_{i,a1}||\boldsymbol{S}_{\bullet,a1}(t_0)||_2]/\Delta(\bar{\boldsymbol{\Upsilon}} + \bar{\boldsymbol{\mathcal{B}}})])$ with $a = x, y, z, \psi$. According to (34) and Assumption 2, it follows that $\|\dot{\boldsymbol{p}}_i\|_\infty \leq \|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d$ is bounded. Then, we can get

$$||\boldsymbol{\mathcal{V}}_i||_\infty = ||\left(\boldsymbol{R}_{i,b}^{i,e}\right)^{\mathrm{T}}\dot{\boldsymbol{p}}_i||_\infty \leq ||\left(\boldsymbol{R}_{i,b}^{i,e}\right)^{\mathrm{T}}||_\infty||\dot{\boldsymbol{p}}_i||_\infty \leq \sqrt{2}||\dot{\boldsymbol{p}}_i||_\infty \leq \sqrt{2}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right). \tag{35}$$

Theorem 1: The LMPC strategy satisfies recursive feasibility for all $t > 0$ if the design parameters satisfy

$$\begin{aligned}\tau_{\max} \geq{}& \sqrt{2}\bar{\gamma}_1\bar{\beta}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right) + \frac{\bar{\gamma}_1\sqrt{2}}{(d_i + b_i)}\\ &\Bigg((d_i + b_i) \times \left(2\sqrt{2}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right)^2 + \bar{G}_i\right) + \sum_{j\in N_i} a_{i,j}\\ &\times \left(\bar{G}_j + 2\sqrt{2}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right)^2 + 2\bar{\beta}\left(\|\mathcal{P}\|_\infty\right.\right.\\ &\left.\left. + \bar{\mathcal{Y}}_d\right) + \sqrt{2}\bar{\gamma}_2||\boldsymbol{\eta}_j||_\infty\right) + b_i\bar{\mathcal{Y}}_d + (\|\boldsymbol{k}_{i,1}\|_\infty\\ &+ \|\boldsymbol{k}_{i,2}\|_\infty)||\boldsymbol{S}_{i,2}||_\infty + (1 + \|\boldsymbol{k}_{i,1}^2\|_\infty)||\boldsymbol{S}_{i,1}||_\infty\Bigg).\end{aligned} \tag{36}$$

Proof: We assume that $||\boldsymbol{\gamma}_i^{-1}||_\infty < \bar{\gamma}_1$, $||\boldsymbol{\gamma}_j||_\infty < \bar{\gamma}_2$ and $||\boldsymbol{\beta}_i||_\infty < \bar{\beta}$ since the system parameters of UAVs are bounded in practice, where $i = 1, 2, \ldots, N$. For the actual auxiliary controller $\boldsymbol{\eta}_i$, we have

$$\begin{aligned}||\boldsymbol{\eta}_i||_\infty \leq{}& ||\boldsymbol{\gamma}_i^{-1}||_\infty||\boldsymbol{\beta}_i||_\infty||\boldsymbol{\mathcal{V}}_i||_\infty + \frac{||\boldsymbol{\gamma}_i^{-1}||_\infty||\left(\boldsymbol{R}_{i,b}^{i,e}(\psi_i)\right)^{\mathrm{T}}||_\infty}{(d_i + b_i)}\\ &\times \Bigg((d_i + b_i)\left(\|\ddot{\boldsymbol{G}}_i\|_\infty + \|\dot{\boldsymbol{R}}_{i,b}^{i,e}(\psi_i)\|_\infty||\boldsymbol{\mathcal{V}}_i||_\infty\right)\\ &+ \sum_{j\in N_i} a_{i,j}\left(\|\ddot{\boldsymbol{G}}_j\|_\infty + ||\dot{\boldsymbol{R}}_{j,b}^{j,e}(\psi_j)||_\infty||\boldsymbol{\mathcal{V}}_j||_\infty\right.\\ &\left. + \sqrt{2}||\dot{\boldsymbol{\mathcal{V}}}_j||_\infty\right) + b_i\bar{\mathcal{Y}}_d + \left(\|\boldsymbol{k}_{i,1}\|_\infty + \|\boldsymbol{k}_{i,2}\|_\infty\right)\\ &\times ||\boldsymbol{S}_{i,2}||_\infty + \left(1 + \|\boldsymbol{k}_{i,1}^2\|_\infty\right)||\boldsymbol{S}_{i,1}||_\infty\Bigg)\\ \leq{}& \sqrt{2}\bar{\gamma}_1\bar{\beta}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right) + \frac{\bar{\gamma}_1\sqrt{2}}{(d_i + b_i)}\\ &\Bigg((d_i + b_i) \times \left(2\sqrt{2}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right)^2 + \bar{G}_i\right) + \sum_{j\in N_i} a_{i,j}\\ &\times \left(\bar{G}_j + 2\sqrt{2}\left(\|\mathcal{P}\|_\infty + \bar{\mathcal{Y}}_d\right)^2 + 2\bar{\beta}\left(\|\mathcal{P}\|_\infty\right.\right.\\ &\left.\left. + \bar{\mathcal{Y}}_d\right) + \sqrt{2}\bar{\gamma}_2||\boldsymbol{\eta}_j||_\infty\right) + b_i\bar{\mathcal{Y}}_d + (\|\boldsymbol{k}_{i,1}\|_\infty\end{aligned}$$

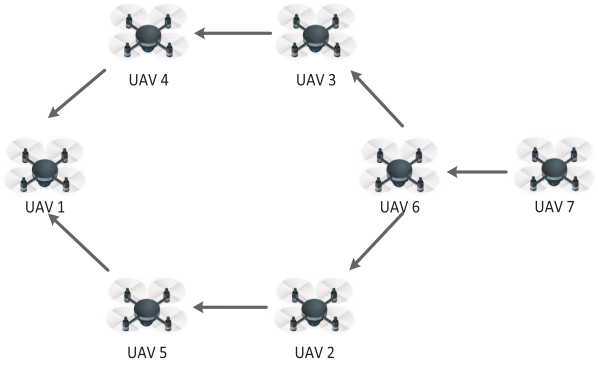


Fig. 1. Communication topology.

$$\begin{aligned}&+ \|\boldsymbol{k}_{i,2}\|_\infty)||\boldsymbol{S}_{i,2}||_\infty + (1 + \|\boldsymbol{k}_{i,1}^2\|_\infty)||\boldsymbol{S}_{i,1}||_\infty\Bigg)\\ &\leq \tau_{\max}\end{aligned} \tag{37}$$

where $\|\dot{\boldsymbol{R}}_{i,b}^{i,e}(\psi_i)\| \leq \sqrt{2}||\boldsymbol{\mathcal{V}}_i||_\infty$, $||\ddot{\boldsymbol{G}}_i||_\infty \leq \bar{G}_i$ and $||\ddot{\boldsymbol{G}}_j||_\infty \leq \bar{G}_j$. Then, we can obtain that the LMPC strategy satisfies recursive feasibility for all $t > 0$ if $\tau_{\max}$ satisfies the inequality (37). The proof is thus completed. ■

B. Stability Analysis

The backstepping controller $\boldsymbol{\eta}_i$ guarantees the feasibility property of the LMPC strategy. The proposed LMPC seeks the optimal solution within the range of feasible solution $\boldsymbol{\eta}_i$.

Lemma 4: By adding the backstepping controller $\boldsymbol{\eta}_i$ to the constraint of (16), the LMPC algorithm guarantees that the closed-loop system is asymptotically stable with a stability region of attraction.

Proof: According to dynamics system of quadrotor UAVs, one has

$$\begin{cases}\hbar_1(||\boldsymbol{\mathcal{X}}_i||) \leq V_{i,2}(\boldsymbol{\mathcal{X}}_i) \leq \hbar_2(||\boldsymbol{\mathcal{X}}_i||)\\ \frac{\partial V_{i,2}(\boldsymbol{\mathcal{X}}_i)}{\partial \boldsymbol{\mathcal{X}}_i}\boldsymbol{g}_i\left(\boldsymbol{\mathcal{X}}_i, \boldsymbol{\eta}_i(\boldsymbol{\mathcal{X}}_i)\right) \leq -\hbar_3(||\boldsymbol{\mathcal{X}}_i||)\end{cases} \tag{38}$$

where $\hbar_1$, $\hbar_2$ and $\hbar_3$ are functions of class $\mathcal{K}$.

Taking (16) into consideration, the optimal solution is selected for each sampling period. Then, one has

$$\frac{\partial V_{i,2}(\boldsymbol{\mathcal{X}}_i)}{\partial \boldsymbol{\mathcal{X}}_i}\boldsymbol{g}_i(\boldsymbol{\mathcal{X}}_i, \boldsymbol{\tau}_i) \leq \frac{\partial V_{i,2}(\boldsymbol{\mathcal{X}}_i)}{\partial \boldsymbol{\mathcal{X}}_i}\boldsymbol{g}_i\left(\boldsymbol{\mathcal{X}}_i, \boldsymbol{\eta}_i\right) \leq -\hbar_3(||\boldsymbol{\mathcal{X}}_i||). \tag{39}$$

According to (39), it can be concluded that the LMPC controller $\boldsymbol{\tau}_i$ guarantees that the closed-loop system is asymptotically stable with a stability region of attraction. ■

V. Simulation Results

To show the effectiveness of the proposed LMPC for practical applications, the performance of the proposed controller and the adaptive backstepping controller is evaluated based on [27]. The communication topology is shown in Fig. 1 with six following UAVs labeled from 1 to 6 and one tracked UAV labeled 7. Table III shows the relevant parameters of the quadrotor UAV. The designed parameters of LMPC are shown in Table IV.

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TABLE III
DEFINITION OF THE UAV PARAMETERS

Parameters	Values	Parameters	Values
L_{ix}	0.83	K_{ix}	1
L_{iy}	0.77	K_{iy}	1
L_{iz}	0.5	K_{iz}	1
$L_{i\psi}$	0.51	$K_{i\psi}$	$\pi/180$

TABLE IV
MODEL PREDICTIVE CONTROLLER DESIGN PARAMETERS

Parameters	Values
Discretization steps	350
The sampling period	0.05
The prediction horizon	0.2
The formation switching time	15

Example 1: We consider the formation tracking-containment control without obstacle avoidance. The trajectory of the tracked UAV is selected as

$$\begin{cases} x_r = 0.5t, \quad y_r = -0.5\cos(0.5t) \\ z_r = 2t, \quad \psi_r = \text{atan2}\{\dot{y}_r(t), \dot{x}_r(t)\} \\ \vartheta_{rx} = \sqrt{\dot{x}_r(t) + \dot{y}_r(t)}, \quad \vartheta_{ry} = 0 \\ \vartheta_{rz} = 2, \quad \vartheta_{r\psi} = \frac{\dot{x}_r\ddot{y}_r - \ddot{x}_r\dot{y}_r}{\dot{x}_r^2 + \dot{y}_r^2}. \end{cases} \tag{40}$$

The relative distances of i and j with respect to the tracked UAV are designed to be

$$M_1 = [-2, 0, 0, 0]^{\mathrm{T}}, \quad M_2 = [-2, 2, 0, 0]^{\mathrm{T}}$$
$$M_3 = [-2, -2, 0, 0]^{\mathrm{T}}, \quad M_4 = [2, 2, 0, 0]^{\mathrm{T}}$$
$$M_5 = [2, 0, 0, 0]^{\mathrm{T}}, \quad M_6 = [2, -2, 0, 0]^{\mathrm{T}}.$$

We choose the design parameters as $\boldsymbol{k}_{i,1} = \text{diag}\{7, 7, 7, 6\}$ and $\boldsymbol{k}_{i,2} = \text{diag}\{2, 2, 2, 2\}$. The weighted matrix $\boldsymbol{Q}_J$ of formation errors is $\boldsymbol{Q}_J = \text{diag}\{2 \times 10^4, 2 \times 10^4, 10^4, 10^2\}$, the weighted matrix $\boldsymbol{F}_J$ of system input is $\boldsymbol{F}_J = \text{diag}\{10^2, 10^2, 10^2, 10^2\}$, $\bar{h}_1 = \bar{h}_2 = \bar{h}_3 = \bar{h}_4 = \bar{h}_5 = \bar{h}_6 = -1.2$, and the maximum of the system input is 100. The shifting functions are chosen as $\boldsymbol{G}_1 = [G_{1x}, 0, 0, 0]^{\mathrm{T}}, \boldsymbol{G}_2 = [G_{2x}, -G_{2y}, 0, 0]^{\mathrm{T}}, \boldsymbol{G}_3 = [G_{3x}, G_{3y}, 0, 0]^{\mathrm{T}}, \boldsymbol{G}_4 = [-G_{4x}, -G_{4y}, 0, 0]^{\mathrm{T}}, \boldsymbol{G}_5 = [-G_{5x}, 0, 0, 0]^{\mathrm{T}}$ and $\boldsymbol{G}_6 = [-G_{6x}, G_{6y}, 0, 0]^{\mathrm{T}}$, $T_{i,a} = 15$ and $\delta_i = 0.2$ with $i = 1, 2, 3, 4, 5, 6$.

The initial condition is set as $\boldsymbol{\mathcal{X}}_{1,1}(0) = [0, 2.5, \pi/2, 0.3]^{\mathrm{T}}, \boldsymbol{\mathcal{X}}_{2,1}(0) = [-0.5, 3, \pi/4, 0.3]^{\mathrm{T}}, \boldsymbol{\mathcal{X}}_{3,1}(0) = [0, 0.8, \pi/2, 0.3]^{\mathrm{T}}, \boldsymbol{\mathcal{X}}_{4,1}(0) = [0, 6, \pi/2, 0.3]^{\mathrm{T}}, \boldsymbol{\mathcal{X}}_{5,1}(0) = [0.5, 5, \pi/2, 0.3]^{\mathrm{T}}, \boldsymbol{\mathcal{X}}_{6,1}(0) = [0.1, 2, \pi/2, 0.3]^{\mathrm{T}}$ and $\boldsymbol{\mathcal{X}}_{i,2}(0) = [0, 0, 0, 0]^{\mathrm{T}}$.

The simulation results are shown in Figs. 2–4. 'BS' in the simulation stands for the abbreviation of backstepping. Fig. 2 displays the formation trajectories of tracking-containment control by the backstepping method and the LMPC method. All the common parameters involved are the same for both methods. It is seen that the swarm system of UAVs performs tracking control for the tracked target in the first stage, and then performs containment control for the tracked target in the second stage. It follows from Figs. 3 and 4 that the proposed controller achieves a faster response by predicting future states. The proposed strategy can achieve faster convergence and more stable formation maintenance distance than the backstepping strategy.

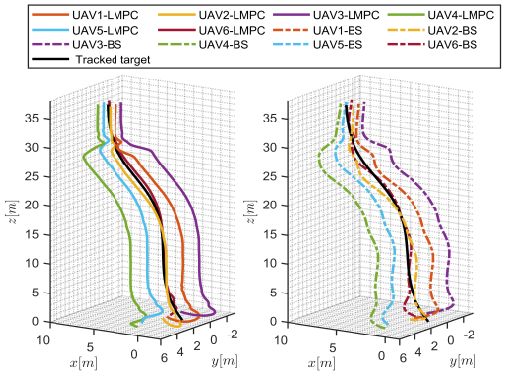


Fig. 2. Formation trajectories of tracking-containment control by the LMPC method and the backstepping method.

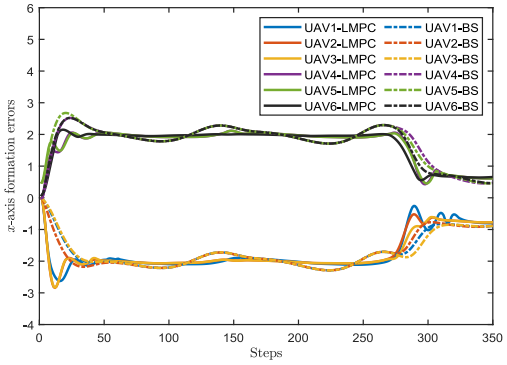


Fig. 3. x-axis formation errors of the tracked target and UAVs.

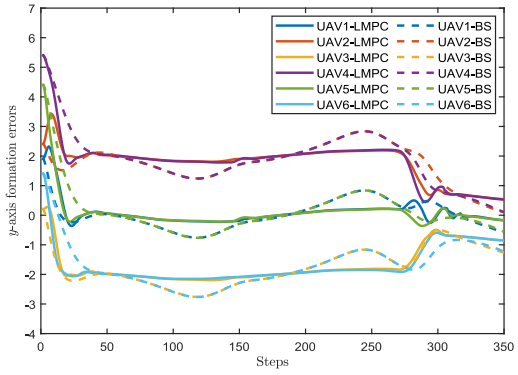


Fig. 4. y-axis formation errors of the tracked target and UAVs.

Example 2: Let us consider a scenario where the swarm system of UAVs crosses multiple obstacles to achieve the formation tracking-containment control and must avoid collision with them. The communication topology and all common design parameters are the

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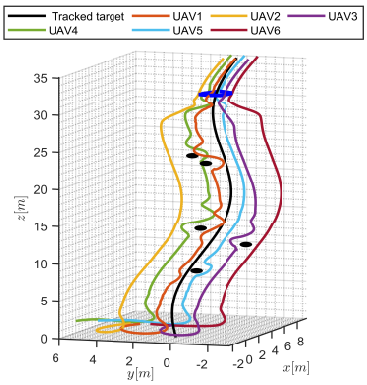


Fig. 5. Formation trajectories of tracking-containment control in scenarios with multiple obstacles.

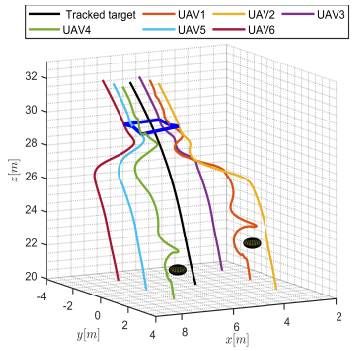


Fig. 6. Trajectories of tracking-containment control.

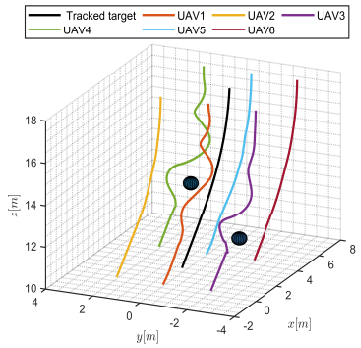


Fig. 7. Obstacle avoidance situation of UAVs.

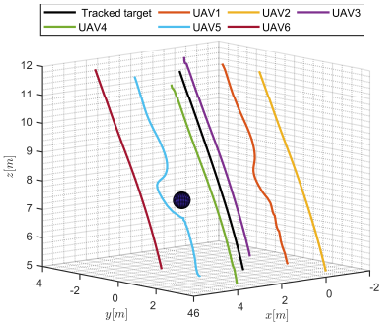


Fig. 8. Obstacle avoidance situation of UAVs.

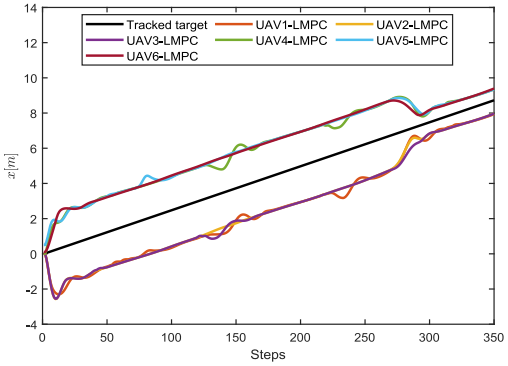


Fig. 9. x-axis tracking trajectories.

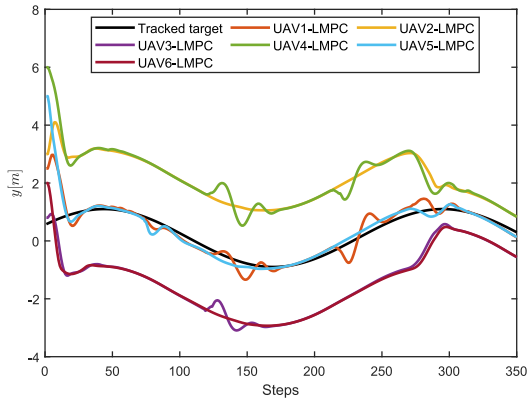


Fig. 10. y-axis tracking trajectories.

same as in Example 1. The obstacle coordinates are set to $[X_{1o}, Y_{1o}, Z_{1o}] = [7.2, 2.2, 22]$, $[X_{2o}, Y_{2o}, Z_{2o}] = [3.2, 0.1, 22.4]$, $[X_{3o}, Y_{3o}, Z_{3o}] = [0.9, -2.9, 12.5]$, $[X_{4o}, Y_{4o}, Z_{4o}] = [5, 1, 13]$, $[X_{5o}, Y_{5o}, Z_{5o}] = [3.8, 0.8, 7.7]$. The parameters $\mathcal{M}_{il}(l = 1, 2, \ldots, 5)$ for obstacle avoidance function are selected as $\mathcal{M}_{i1} = \mathcal{M}_{i2} = \mathcal{M}_{i3} = \mathcal{M}_{i4} = \mathcal{M}_{i5} = 50\,000$, $\wp_{ij} = 1000$, $\bar{D}_{il}$=1, $\mathcal{L}_l = 0.4$ and $\bar{\mathcal{D}}_{ij} = 0.3$.

The simulation results are shown in Figs. 5–13. Fig. 5 depicts the formation trajectories of tracking-containment control with obstacle avoidance. Fig. 6 illustrates the details of the tracking-control transition trajectory. From Figs. 7 and 8, it can be seen that the proposed LMPC strategy can find the optimal paths without crossing multiple obstacles. In Fig. 9, the solid line shows the x-axis tracking trajectories of tracking-containment control under the LMPC method, and the chain line shows the x-axis tracking trajectories of tracking-containment control under the backstepping method. Similarly, Figs. 10 and 11 show the y-axis and z-axis trajectories, respectively. It is worth noting that the trajectory jitter of the LMPC strategy in Figs. 9 and 10 is caused by obstacle avoidance. Compared with the backstepping control strategy, the LMPC strategy can systematically solve the obstacle avoidance problem. The proposed controller achieves faster response and better-tracking performance by predicting future states. Fig. 12 and Fig. 13 show the control input signals $\boldsymbol{\tau}_i$ (τ_{ix}, τ_{iy}, τ_{iz}, and $\tau_{i\psi}$), and the red lines are the constraint

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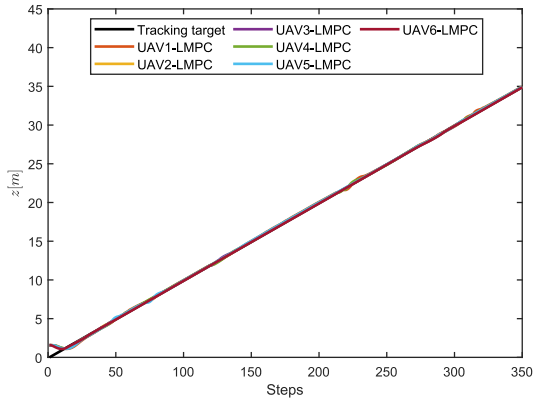


Fig. 11. z-axis tracking trajectories.

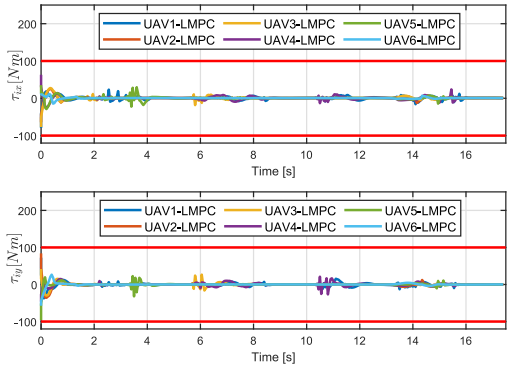


Fig. 12. Control input signals τ_{ix} and τ_{iy} of UAVs.

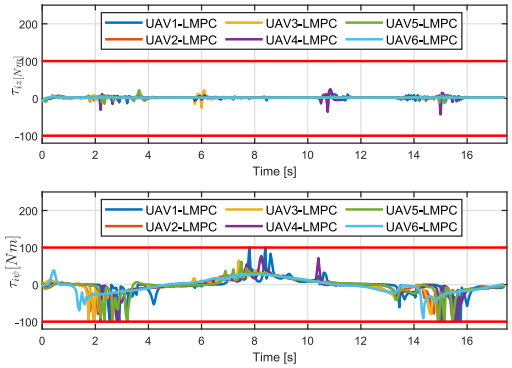


Fig. 13. Control input signals τ_{iz} and $\tau_{i\psi}$ of UAVs.

boundaries. It follows from Fig. 12 and Fig. 13 that the control inputs do not exceed the constraint boundaries.

VI. Conclusion

In this article, a model predictive formation tracking-containment control algorithm for the swarm system of UAVs with obstacle avoidance has been proposed. By introducing a shifting function, the swarm system of UAVs has realized the shifting process from tracking formation to containment formation. The swarm system of UAVs can achieve 3-D collision avoidance in scenarios with multiple obstacles. The proposed LMPC algorithm inherits the stability properties of the backstepping controller and seeks the optimal solution within the range of feasible solution of backstepping. Finally, the simulation has proved the feasibility and stability of the proposed controller. In the future, how to extend the fuzzy LMPC strategy for heterogeneous multiagent systems with dynamic obstacle avoidance is our research interest.

References

[1] M. Ryll, H. H. Bülthoff, and P. R. Giordano, "A novel overactuated quadrotor unmanned aerial vehicle: Modeling, control, and experimental validation," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 2, pp. 540–556, Mar. 2015.

[2] X. Zhou et al., "Safety flight control design of a quadrotor UAV with capability analysis," *IEEE Trans. Cybern.*, vol. 53, no. 3, pp. 1738–1751, Mar. 2023.

[3] A. Abbaspour, K. K. Yen, P. Forouzannezhad, and A. Sargolzaei, "A neural adaptive approach for active fault-tolerant control design in UAV," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 50, no. 9, pp. 3401–3411, Sep. 2020.

[4] F. Chen, R. Jiang, K. Zhang, B. Jiang, and G. Tao, "Robust backstepping sliding-mode control and observer-based fault estimation for a quadrotor UAV," *IEEE Trans. Ind. Electron.*, vol. 63, no. 8, pp. 5044–5056, Aug. 2016.

[5] J. Moreno-Valenzuela, R. Pérez-Alcocer, M. Guerrero-Medina, and A. Dzul, "Nonlinear PID-type controller for quadrotor trajectory tracking," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 5, pp. 2436–2447, Oct. 2018.

[6] D. Wang, Q. Pan, Y. Shi, J. Hu, and C. Zhao, "Efficient nonlinear model predictive control for quadrotor trajectory tracking: Algorithms and experiment," *IEEE Trans. Cybern.*, vol. 51, no. 10, pp. 5057–5068, Oct. 2021.

[7] H. Ren, H. Ma, H. Li, and Z. Wang, "Adaptive fixed-time control of nonlinear MASs with actuator faults," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 5, pp. 1252–1262, May 2023.

[8] J. Berberich, J. Köhler, M. A. Muller, and F. Allgower, "Linear tracking MPC for nonlinear systems part I: The model-based case," *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4390–4405, Sep. 2022.

[9] K. Zhang, Y. Shi, and H. Sheng, "Robust nonlinear model predictive control based visual servoing of quadrotor UAVs," *IEEE/ASME Trans. Mechatronics*, vol. 26, no. 2, pp. 700–708, Apr. 2021.

[10] A. Liu, W.-A. Zhang, L. Yu, H. Yan, and R. Zhang, "Formation control of multiple mobile robots incorporating an extended state observer and distributed model predictive approach," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 50, no. 11, pp. 4587–4597, Nov. 2020.

[11] H. Zhang, H. Zhang, Z. Wang, C. Huang, and Y. Li, "Adaptive event-based predictive lateral following control for unmanned ground vehicle system," *Int. J. Robust Nonlin. Control*, vol. 31, no. 10, pp. 4744–4763, 2021.

[12] P. D. Christofides, J. Liu, and D. M. De La Pena, *Networked and Distributed Predictive Control: Methods and Nonlinear Process Network Applications*. Berlin, Germany: Springer, 2011.

[13] D. N. Bui, T. T. Van Nguyen, and M. D. Phung, "Lyapunov-based nonlinear model predictive control for attitude trajectory tracking of unmanned aerial vehicles," *Int. J. Aeronaut. Space Sci.*, vol. 24, no. 2, pp. 502–513, 2023.

[14] C. Shen, Y. Shi, and B. Buckham, "Trajectory tracking control of an autonomous underwater vehicle using Lyapunov-based model predictive control," *IEEE Trans. Ind. Electron.*, vol. 65, no. 7, pp. 5796–5805, Jul. 2018.

[15] X. Yu, N. Ding, A. Zhang, and H. Qian, "Cooperative moving-target enclosing of networked vehicles with constant linear velocities," *IEEE Trans. Cybern.*, vol. 50, no. 2, pp. 798–809, Feb. 2020.

[16] X. Dong, Y. Zhou, Z. Ren, and Y. Zhong, "Time-varying formation tracking for second-order multi-agent systems subjected to switching topologies with application to quadrotor formation flying," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 5014–5024, Jun. 2017.

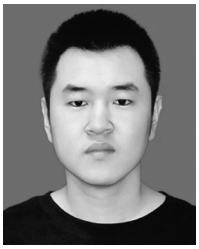
[17] Y. Gao, C. Bai, L. Zhang, and Q. Quan, "Multi-UAV cooperative target encirclement within an annular virtual tube," *Aerosp. Sci. Technol.*, vol. 128, Sep. 2022, Art. no. 107800.

[18] L. Kou, Z. Chen, and J. Xiang, "Cooperative fencing control of multiple vehicles for a moving target with an unknown velocity," *IEEE Trans. Autom. Control*, vol. 67, no. 2, pp. 1008–1015, Feb. 2022.

[19] L. Kou, Y. Huang, Z. Chen, S. He, and J. Xiang, "Cooperative fencing control of multiple second-order vehicles for a moving target with and without velocity measurements," *Int. J. Robust Nonlin. Control*, vol. 31, no. 10, pp. 4602–4615, 2021.

Authorized licensed use limited to: Wuhan University. Downloaded on August 22,2025 at 08:14:57 UTC from IEEE Xplore. Restrictions apply.

[20] D. Yu, J. Long, C. L. P. Chen, and Z. Wang, "Bionic tracking-containment control based on smooth transition in communication," *Inf. Sci.*, vol. 587, pp. 393–407, Mar. 2022.
[21] Q. Quan, R. Fu, M. Li, D. Wei, Y. Gao, and K.-Y. Cai, "Practical distributed control for VTOL UAVs to pass a virtual tube," *IEEE Trans. Intell. Veh.*, vol. 7, no. 2, pp. 342–353, Jun. 2022.
[22] C. Mu and J. Peng, "Learning-based cooperative multiagent formation control with collision avoidance," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 12, pp. 7341–7352, Dec. 2022.
[23] X. Wang and D. Ye, "Finite-time output-feedback formation control for high-order nonlinear multiagent systems with obstacle avoidance," *IEEE Trans. Autom. Sci. Eng.*, early access, Feb. 22, 2023, doi: 10.1109/TASE.2023.3244536.
[24] Z. Peng, D. Wang, T. Li, and M. Han, "Output-feedback cooperative formation maneuvering of autonomous surface vehicles with connectivity preservation and collision avoidance," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2527–2535, Jun. 2020.
[25] R. Fu, Q. Quan, M. Li, and K.-Y. Cai, "Practical distributed control for cooperative multicopters in structured free flight concepts," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4203–4216, Apr. 2023.
[26] C. K. Verginis and D. V. Dimarogonas, "Adaptive robot navigation with collision avoidance subject to 2nd-order uncertain dynamics," *Automatica*, vol. 123, Jan. 2021, Art. no. 109303.
[27] H. Wei, C. Shen, and Y. Shi, "Distributed Lyapunov-based model predictive formation tracking control for autonomous underwater vehicles subject to disturbances," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 51, no. 8, pp. 5198–5208, Aug. 2021.
[28] C. Zhao, D. Wang, J. Hu, and Q. Pan, "Nonlinear model predictive control-based guidance algorithm for quadrotor trajectory tracking with obstacle avoidance," *J. Syst. Sci. Complex.*, vol. 34, no. 4, pp. 1379–1400, 2021.
[29] J. Dentler et al., "Collision avoidance effects on the mobility of a UAV swarm using chaotic ant colony with model predictive control," *J. Intell. Robot. Syst.*, vol. 93, no. 1, pp. 227–243, 2019.
[30] M. Castillo-Lopez, M. A. Olivares-Mendez, and H. Voos, "Evasive maneuvering for UAVs: An MPC approach," in *Proc. 3rd Iberian Robot. Conf. ROBOT*, 2018, pp. 829–840.
[31] M. Castillo-Lopez, S. A. Sajadi-Alamdari, J. L. Sanchez-Lopez, M. A. Olivares-Mendez, and H. Voos, "Model predictive control for aerial collision avoidance in dynamic environments," in *Proc. 26th Mediterr. Conf. Control Autom. (MED)*, 2018, pp. 1–6, doi: 10.1109/MED.2018.8442967.
[32] G. Lin, H. Li, C. K. Ahn, and D. Yao, "Event-based finite-time neural control for human-in-the-loop UAV attitude systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10387–10397, Dec. 2023.
[33] H. Liang, Z. Du, T. Huang, and Y. Pan, "Neuroadaptive performance guaranteed control for multiagent systems with power integrators and unknown measurement sensitivity," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 9771–9782, Dec. 2023.
[34] P. Du, W. Zhong, X. Peng, L. Li, and Z. Li, "Self-healing control for wastewater treatment process based on variable-gain state observer," *IEEE Trans. Ind. Informat.*, vol. 19, no. 10, pp. 10412–10424, Oct. 2023.
[35] H. Ren, Z. Cheng, J. Qin, and R. Lu, "Deception attacks on event-triggered distributed consensus estimation for nonlinear systems," *Automatica*, vol. 154, Aug. 2023, Art. no. 111100.



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